WEEK 9

Data Science II

OULAD: cutoff day, course-history features and temporal split

Bera Ergin · s29315 · PJAIT · Türksat internship

Goal of this week

1

Data

OULAD downloaded and its sha256 pinned. The 10.6-million-row click table was read in chunks and reduced to a daily summary.

2

Features

Day-60 cutoff, at_risk target, general and course-history features, a leakage-free temporal split.

3

Notebook

02_oulad_eda (TR/EN) and HTML. Numbers come only from cell outputs.

No model was trained this week. The general vs course-history model comparison comes in Week 10.

Why OULAD?

32,593	7 / 22	206	10,655,280
records (student × presentation)	modules / presentations	assessments and exams	click rows

Source: Kuzilek, Hlosta & Zdrahal (2017), Open University Learning Analytics Dataset, UCI (id 349), CC BY 4.0. Two things UCI lacks are present here: several runs of the same course (2013B, 2013J, 2014B, 2014J) and dated assessment and activity records. This makes it possible to test the course-context hypothesis and to define a real early-warning moment.

Data and verification

Table	Rows	Note
studentInfo	32,593	Key unique; imd_band 1,111 missing
studentRegistration	32,593	date_unregistration 22,521 empty (still enrolled)
assessments	206	date 11 missing
studentAssessment	173,912	score 173 missing
vle / courses	6,364 / 22	—
studentVle	10,655,280	Read in chunks

sha256

Archive digest pinned; verified on every run.

Chunked summary

Raw table and daily summary match exactly in rows (10,655,280) and total clicks (39,605,099).

Early-warning moment and target

```
at_risk = final_result ∈ {Fail, Withdrawn}
```

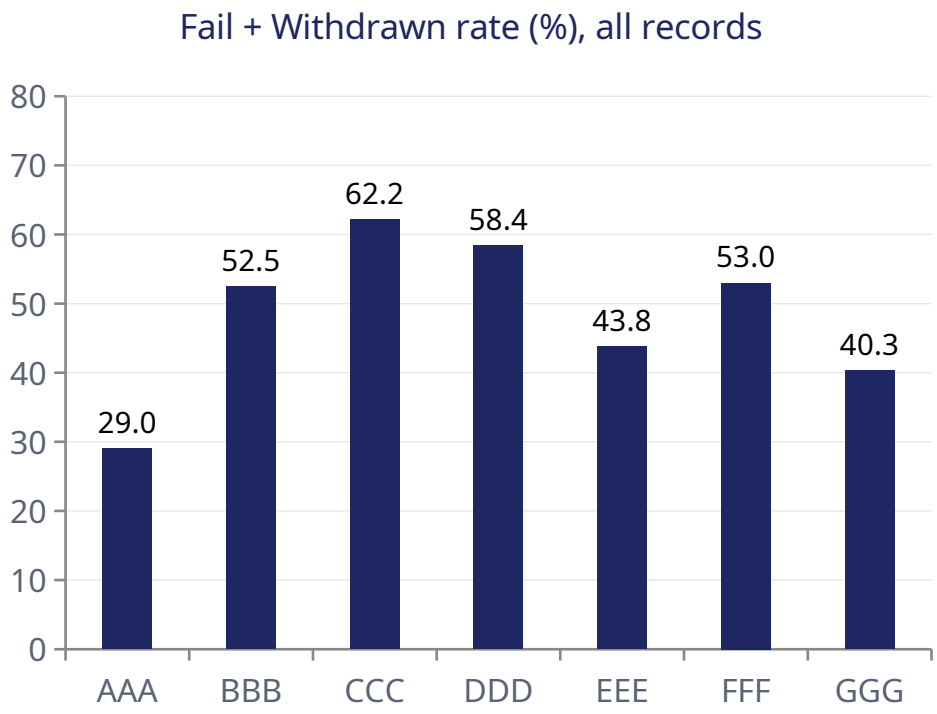
- Cutoff day: CUTOFF_DAY = 60 (configurable).
- Only assessments dated and submitted by day 60; final exam excluded.
- Clicks up to day 60.
- Students unregistered before day 60 are excluded; later ones stay with at_risk = 1.

32,593 all records

−6,190 left before cutoff

26,403 sample

Course context: large gaps between modules



- Across modules the rate ranges from 29.0% (AAA) to 62.2% (CCC).
- Runs of the same module are close: at most 6.5 points apart (EEE).
- In the sample (after day 60), lowest is AAA 2013J at 23.4%, highest DDD 2013B at 52.6%.
- Descriptive finding: a course's past risk rate appears informative. The evidence comes from the Week 10 comparison.

Course-history features

For presentation P, history = ONLY strictly earlier presentations of the same module. P and later never enter any feature.

Feature	Definition
`hist_slot_mean`	Past mean score of the same assessment slot (module, type, order): its difficulty
`hist_slot_submit_rate`	Past submission rate of the same slot
`score_vs_hist`	Student's score − hist_slot_mean (student average)
`hist_module_risk_rate`	at_risk rate of earlier presentations (same sample definition)
`has_history`	0 if no earlier presentation; history columns stay NaN, not imputed

General features (up to day 60)

Assessments

- Expected, submitted and unsubmitted counts
- Weighted and unweighted mean score
- Mean delay (submitted − due)
- is_banked row count (rows kept)

Activity (VLE)

- Total clicks, active days
- Days from last activity to cutoff
- Clicks in the 5 most frequent activity types
- date_registration

Not used: sociocultural columns (gender, region, education, imd_band, age, disability), num_of_prev_attempts, studied_credits. 26 columns in total.

Temporal split

Split	Presentations	Rows	at_risk	Has history
history	2013B	3,927	46.7%	0.0%
train	2013J + 2014B	13,639	42.6%	74.1%
test	2014J	8,837	38.3%	100.0%

- 2013B is a history source only; it enters neither training nor test.
- Test is 100% with history: 2014J has an earlier run in every module.
- Train rows without history: the first run (2013J) of AAA, EEE, GGG and CCC 2014B; these modules have no 2013B.

Missing values are consistent

Column	Missing	Explanation
`hist_module_risk_rate`	7,465	= history 3,927 + train without history 3,538. Expected.
`hist_slot_mean`	8,952	7,465 + 1,487 (GGG, see data note 1)
`score_vs_hist`	10,046	Needs both history and a student score
`score_mean`	3,995	No graded submission by day 60
`score_weighted_mean`	4,172	score_mean + 177 (BBB, see data note 2)
`vle_days_to_cutoff`	564	No activity at all by day 60
`date_registration`	7	Missing in the source table

Data notes (not bugs)

1 · GGG 2014B / 2014J

All 1,487 rows have n_expected = 0 (790 + 697). These runs have no graded assessment by day 60, so there is no slot to match.

Result: GGG has only the activity signal at the cutoff.

2 · BBB 2014J

A TMA with weight 0 falls before day 60. For the 177 students who submitted only that one, the weighted mean is undefined.

Result: NaN is correct; the unweighted mean exists.

Bug record 1: test environment

Field	Description
Symptom	pytest stopped at collection: ModuleNotFoundError: sklearn. Same error in the Week 8 folder.
Root cause	scikit-learn was not installed in the shared virtual environment used by all weeks. Not caused by code.
Fix	pip install -r requirements.txt in the shared environment.
Status	Verified: 237 passed (after 9a), 243 passed (after 9b).

Mapping to our system

OULAD feature	Our counterpart	Status
`score_mean`	Grade average	**Available**
`vle_* (activity)`	Unexcused absence (approximate)	**Approximate**
`n_unsubmitted`	Missed exam	**To verify**
`delay_mean`	—	**Missing**
`hist_*`	Past-term grades	**Synthetic (W10)**
`Sociocultural`	—	**Not used**

The system has no past-term data. In Week 10 a separate synthetic school history will be generated without touching the mock data. It only shows that "the system works"; the evidence for the hypothesis comes from OULAD.

Notebooks

`02_oulad_eda.ipynb`	Turkish narrative
`02_oulad_eda_eng.ipynb`	English narrative
`notebooks/html/`	HTML exports

Notebooks were executed from scratch and saved with outputs. No numbers in markdown; every figure has a description cell below it. The final OZET cell matches the validate output exactly.

a. Goal
b. Data
c. Sample
d. Split
e. Course context
f. Early signals
g. Data notes
h. Bridge to the school system
i. Limitations
j. Summary

Verification

243 passed

1 warning (StarletteDeprecationWarning)

```
python -m data_science.oulad.download
python -m data_science.oulad.vle
python -m data_science.oulad.dataset
python -m data_science.oulad.validate --
features
```

- OULAD tests use no network; they run on tiny fake tables (9a: 4, 9b: 6 tests).
- Checks: post-cutoff assessments and clicks excluded, unregistration rule, no history leakage, NaN when has_history = 0, final exam excluded, chunked summary = one-pass summary.
- Both notebooks executed from scratch via nbconvert; 0 error cells.
- Application code, mock data and eval were out of scope.

Open items and Week 10

Open items

- Bug record 1 (test environment): verified.
- pytest warning identified (StarletteDeprecationWarning, httpx); not yet resolved.
- Alternative text: 6 images in 01, 7 in 02_eng.
- Week 6 bug 4 and Week 7 m5, c4: open.

Week 10 — Model

- General vs course-history model, same test (2014J)
- Uncertainty intervals and SHAP explanations
- Feature subset compatible with the system
- Synthetic school-history generator (mock data untouched)